

Federico Brancasi

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EXPERIENCE

Research Engineer

February 2026 – Present

Microsoft

Zurich, Switzerland

- Fine-tuning language models and optimizing prompts for VS Code's AI features, reducing Time-To-First-Token (TTFT) and improving response quality for inline code completion.
- Building data pipelines for model fine-tuning, including data collection, cleaning and evaluation workflows to enhance the developer experience.

Machine Learning Researcher

August 2025 – February 2026

CERN

Geneva, Switzerland

- Developed QuantDiff, an automated mixed-precision quantization framework that finds the optimal bit allocation by balancing each layer's sensitivity against its computational cost, enabling precise model compression.
- Created Neural Anomaly Metric, a new evaluation metric that better captures how quantization affects model accuracy, addressing limitations of existing metrics.

Machine Learning Researcher

January 2025 – July 2025

ETH Zurich

Zurich, Switzerland

- Built a custom backend for AMD's Brevitas (PyTorch Quantization Framework) to enable automated neural network quantization and deployment on low-power embedded processors.
- Conducted comprehensive benchmarking of different quantization schemes, with particular attention to non-linear operator quantization and automatic network optimization for embedded deployments.

Research Assistant

February 2023 – July 2024

University of Trento

Trento, Italy

- Modernized Computational Logic course materials and led the development of a new textbook, integrating Knowledge Graph methodologies and creating advanced visualizations with LaTeX and TikZ, resulting in a 30% increase in student pass rates.

EDUCATION

ETH Zurich

Master in Computer Science

Zurich, Switzerland

September 2023 – January 2026

University of Trento

Master in Computer Science

Trento, Italy

September 2023 – January 2026

Aalto University

Master in Computer Science

Helsinki, Finland

September 2023 – January 2026

Eötvös Loránd University

Master in Computer Science

Budapest, Hungary

September 2023 – January 2026

University of Trento

Bachelor in Computer Science

Trento, Italy

September 2020 – September 2023

AWARDS AND CERTIFICATIONS

Anthropic's Claude 4 Hackathon London Winner | [Link](#)

June 2025

- Selected as winner at Anthropic's inaugural student hackathon at the London School of Economics and Political Science (LSE), building an intelligent security system with natural language control. Developed a full-stack application combining real-time video monitoring with Claude's conversational AI capabilities in just an hour.

AI Hackathon Munich Winner | [Link](#)

August 2025

- Won the prototyping sprint at CDTM Munich (Organized by AWS, OpenAI, Lovable, n8n, ElevenLabs, TUM.ai), creating a complete application that transforms any data source (YouTube videos, websites, PDFs, CSVs, etc.) into professional business reports and presentations with one-click AI automation in just 6 hours, demonstrating end-to-end product development from concept to working prototype.

AWS MLOps (Machine Learning Operations) Engineer Path | [Link](#)

July 2024 – October 2024

- Obtained AWS Certified Cloud Practitioner, AWS Certified AI Practitioner and AWS Certified Solutions Architect, currently pursuing additional AWS certifications related to Machine Learning and Cloud Computing.

2nd Place at Startup Lab Competition | [Link](#)

February 2024 – June 2024

- Secured 30K euros pre-incubation funding for an innovative agritech solution, developing an autonomous harvesting prototype that combines YOLO object detection, ROS and SLAM for precision agriculture.

Industrial AI Challenge Winner | [Link](#)

September 2023 – December 2023

- Led the development of an award-winning AI solution for BLM Group's inventory management, implementing machine learning models that reduced waste by 60 percent, improving overall supply chain efficiency.

PROJECTS

DeepQuant Neural Network Quantization Framework | [Link](#)

January 2025 – July 2025

- Led the development a comprehensive library for converting quantized models with AMD's Brevitas library to true integer-only representations, enabling efficient deployment on embedded systems without floating-point units.
- Implemented advanced graph transformation algorithms including quantization node splitting, dequantization unification and custom forward pass injection for neural network layers, supporting complex architectures like CNNs such as ResNet and Transformer-based models including Vision Transformers.

Neural Anomaly Metric | [Link](#)

August 2025 – February 2026

- Developed a new evaluation metric that better measures how quantization affects model accuracy, using anomaly detection to identify quality degradation that existing metrics fail to capture.
- Implemented support for multiple neural network backbones (Vision Transformers, ResNets, EfficientNet), enabling per-image quality assessment with statistical validation across different compression levels.

QuantDiff Mixed-Precision Framework | [Link](#)

August 2025 – February 2026

- Built an automated mixed-precision quantization framework that optimally allocates bits across neural network layers by balancing each layer's sensitivity against its computational cost.
- Enabled precise model compression to any target size budget, supporting 4, 8, and 16-bit precision levels with quality assessment using FID and CLIP metrics.

Deep Learning Study Book | [Link](#)

February 2024 – February 2025

- Authored a comprehensive deep learning textbook using LaTeX and TikZ, creating detailed technical illustrations of neural architectures and advanced ML concepts, publicly available in the University of Trento's shared resources.
- Developed extensive documentation of deep learning principles, including custom visualizations of transformers, CNN architectures and optimization techniques, serving as a key learning resource for future students.

Enhanced File Explorer for Chrome | [Link](#)

September 2024 – February 2026

- Developed a browser extension to enhance local file browsing experience across Chromium browsers.
- Gathered more than 1000 daily users and surpassing 5000 installations on Chrome Web Store.

SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, R, SQL, HTML/CSS, LaTeX, Bash

AI/ML: PyTorch, TensorFlow, LLMs, Transformers, Vision Transformers (ViT), ResNet/CNNs, GPT Architectures, Diffusion Models, Neural Network Quantization, Fine-tuning, Prompt Engineering, RAG, Generative AI, NLP, Scikit-learn, Computer Vision, Model Compression, AI Agents, Anthropic Claude API, OpenAI API, Open Source Models (Llama, Qwen, Gemma, DeepSeek, Phi), Ollama, LM Studio

Frameworks & Tools: FastAPI, React, Node.js, Flask, n8n, LangChain, Hugging Face, Lovable, Streamlit, Gradio, Tailwind CSS, Docker, Git, AWS, MLOps, CI/CD, GitHub Actions

Data Science: Pandas, NumPy, Jupyter, Matplotlib, Seaborn, Plotly, Statistical Analysis, Data Visualization

Developer Tools: VS Code, PyCharm, Linux, GitHub, Google Colab, Cursor, Claude Code, TensorBoard, pytest

Cloud & Infrastructure: AWS (Certified Solutions Architect, AI Practitioner), Cloud Computing, Containerization, System Architecture, Vector Databases (Pinecone, Chroma), Vercel, Netlify, Supabase

APIs & Integration: REST APIs, WebSockets, Zapier, MongoDB, PostgreSQL

Specialized: Hardware Optimization, ONNX, Model Deployment, API Design, Technical Writing, Research Methodology, Workflow Automation